

Intro to Operating Systems

CISC 3310 — CUNY Brooklyn College

Lecture Notes



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Definition of an Operating System

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What is an **Operating System (OS)**?

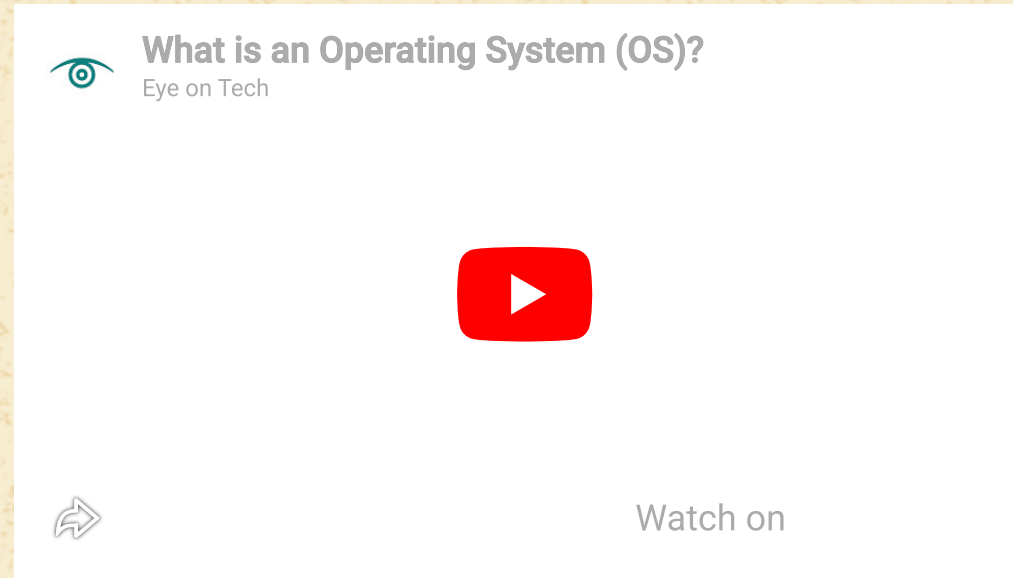


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Definition of an Operating System

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The video at <https://www.youtube.com/watch?v=RhHMgkUdhdk> defines an Operating System in one way:



<https://www.youtube.com/watch?v=RhHMgkUdhdk>



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Definition of an Operating System

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Follow-up questions:

1. What is the definition of OS that the video mentions?
2. How do programs communicate with the OS (e.g., to request services or info?)
3. What are some famous computer operating system brand names?
4. Where are OSs usually installed (besides on computers and phones?)
5. Can you think of additional definitions of OS?



Definition of an Operating System

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Several More Definitions of an OS:

- "A program that manages a computer's resources, especially the allocation of those resources among other programs." — [Encyclopedia Britannica](#)
- "An interface between a computer user and computer hardware." — [Tutorials Point](#)
- "Software that controls the operation of a computer and directs the processing of programs." — [Merriam-Webster](#)
- Another idea: A program that operates from a device's launch to its shutdown.



Reasons for Studying Operating Systems

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Why do we study operating systems? Why wouldn't learning programming be sufficient?

- At any point during your career, you might be asked to analyze or modify the functionality of an operating system.
- You can make your programs work more efficiently by understanding how an operating system executes them.
- The algorithmic approaches that operating systems use to solve various problems may inspire your own programming practice.



Software Types

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To understand how operating systems relate to other programs, we should first learn what software types exist.

1. **System software** is a collection of programs that sit on top of, control, and manage hardware. It provides services to other applications. System software includes:
 - i. The **operating system**, and
 - ii. **Utility software**, which is not as essential as the OS but helps in analyzing the computer's behavior, debugging errors, and optimizing its operation.



Software Types

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2. **Application software** (or **user application software**) includes any applications that let users complete daily tasks. Application software depends on and cannot operate without the services of the system software.

Note, however, that it is sometimes hard to tell whether a piece of software is system software or application software, and there are big debates and even court case decisions on this question. Example: in the [U.S. vs. Microsoft Corp.](#) case, the debate is whether a browser is counted as system software or not.



Software Types

Examples of software included in the **operating system**:

- **Kernel** = a core that includes instructions for the OS's basic functionality.
- **Drivers** = each one provides services for and controls a specific type of hardware device.
- **User Interface** = a means by which a user can interact with the computer: either graphically by clicking on the screen or textually by typing instructions into a terminal/command-line utility.



Software Types

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Examples of software included in **utility software**:

- **Text editors** = provide access to view or edit code and files.
- **Debuggers** = help the programmer detect bugs and analyze a program's execution at any stage.
- **Antivirus programs** = search for virus-like patterns in executables and remove possible threats to the device's operation.
- **File management programs** = allow viewing properties of, organizing, and deleting files.



Software Types

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Examples of programs included in **application software**:

- **Office productivity apps:** [Microsoft Office](#), [LibreOffice](#), [Google's G Suite](#), etc.
- **Gaming apps:** [Minecraft](#), [Gold Miner](#), [Bertie the Brain](#), etc.
- **Music apps:** [Spotify](#), [Pandora](#), [YouTube Music](#), etc.
- **Video conference apps:** [Zoom](#), [Skype](#), [Class Collaborate](#), [Microsoft Teams](#), etc.
- **Emails transfer apps:** [Gmail](#), [Yahoo! Mail](#), [Thunderbird](#), [Microsoft Outlook](#), etc.



Software Types

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Any questions so far?



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Operating System Brand Examples

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Laptops/Desktops:

- Windows
- Mac OS
- Chrome OS
- Linux/UNIX

Phones/Tablets/Smart Watches:

- Android
- iOS

Every electronic device or appliance contains an operating system, no matter how simple its functionality is.



Types of Operating Systems

We can categorize OSs into types based on their functionality patterns:

Types of Operating Systems as Fast As Possible



https://www.youtube.com/watch?v=MR2ntdZW__A



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Types of Operating Systems

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Another video:

types of operating system (OS) | operating system[E...



<https://www.youtube.com/watch?v=SH64-xkxF5w>



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Types of Operating Systems

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Follow-up questions:

1. What are some OS types mentioned in the videos?
2. Give an example of one of the mentioned OS types.
3. What type of OS should be used in medical or military conditions?
4. What type of OS should be used when sharing resources is wanted?



Types of Operating Systems

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Any questions so far?



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Security & Protection

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To ensure that the device functions correctly and benignly, and to avoid the corruption of data, the operating system must reinforce security & protection measures.

Security is the ability of the computer to protect data from malicious actions, usually from outside.

An example of a security measure is to prevent an unauthorized user from accessing the data of a legit user, to which the unauthorized one has no access permissions.



Security & Protection

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Protection is the ability of the computer to protect data from malicious actions, either advertent or unintentional, usually from the inside.

An example of a protection measure is to prevent the data of Program A to be overwritten by Program B due to miscalculations of memory and variable sizes.

We will elaborate on security & protection in one or two lectures at the end of the course.



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Privileges

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Some of the services that a program requests from the operating systems require special permission levels.

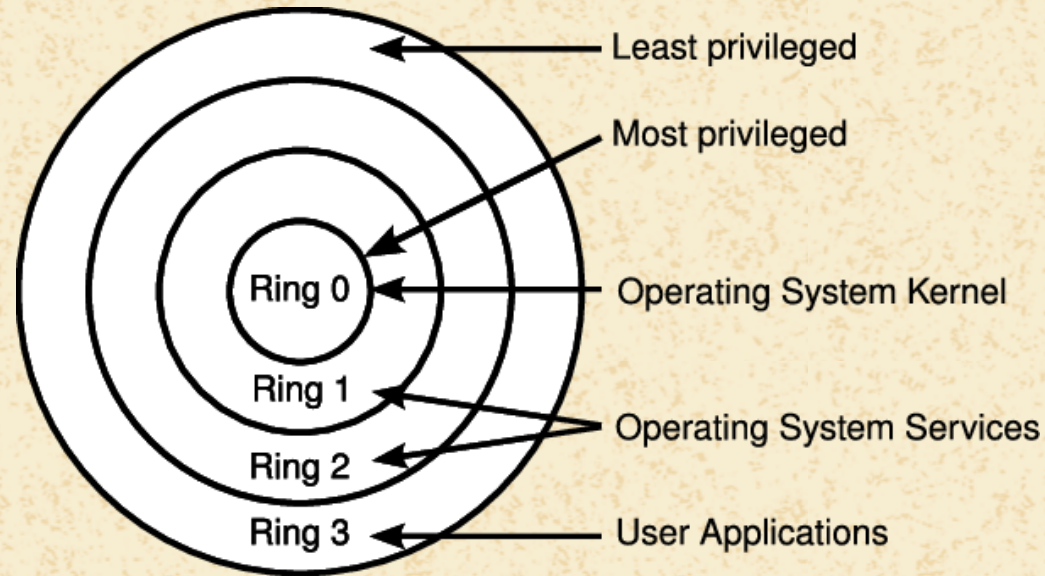
For instance, installing new utilities (programs) on a Windows or Linux device requires administrator permission.

The operating system maintains lists (or matrices) of activities or access types allowed by various users. These tasks are divided into groups, called **privilege rings**, a method that protects the computer and its data from corruption or malfunction.



Privileges

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Taken from https://www.researchgate.net/figure/The-x86-privilege-level-architecture_fig1_235953591.



Privileges

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Any questions so far?



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Intro to Operating Systems: Sources

Bell, John T. [Introduction](#). *University of Illinois, Chicago*.

Wienand, Ian. (2019). [Chapter 4](#). *Computer Science from the Bottom Up*. License: [CC BY-SA: Attribution-ShareAlike](#).

Poluru, Lalitha. [Chapter 2](#). *Information Communication Technology for Libraries*. INFLIBNET Centre, 24 July 2018. License: [CC BY-NC: Attribution-NonCommercial](#).



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